BartsMS

# Epidemiology and Prognosis of MS

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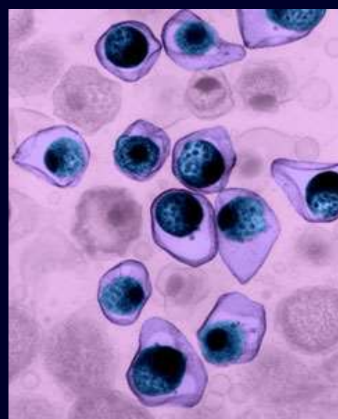
# Potential Triggers for Multiple Sclerosis



Genetic predisposition

Infectious agent

Environmental factors

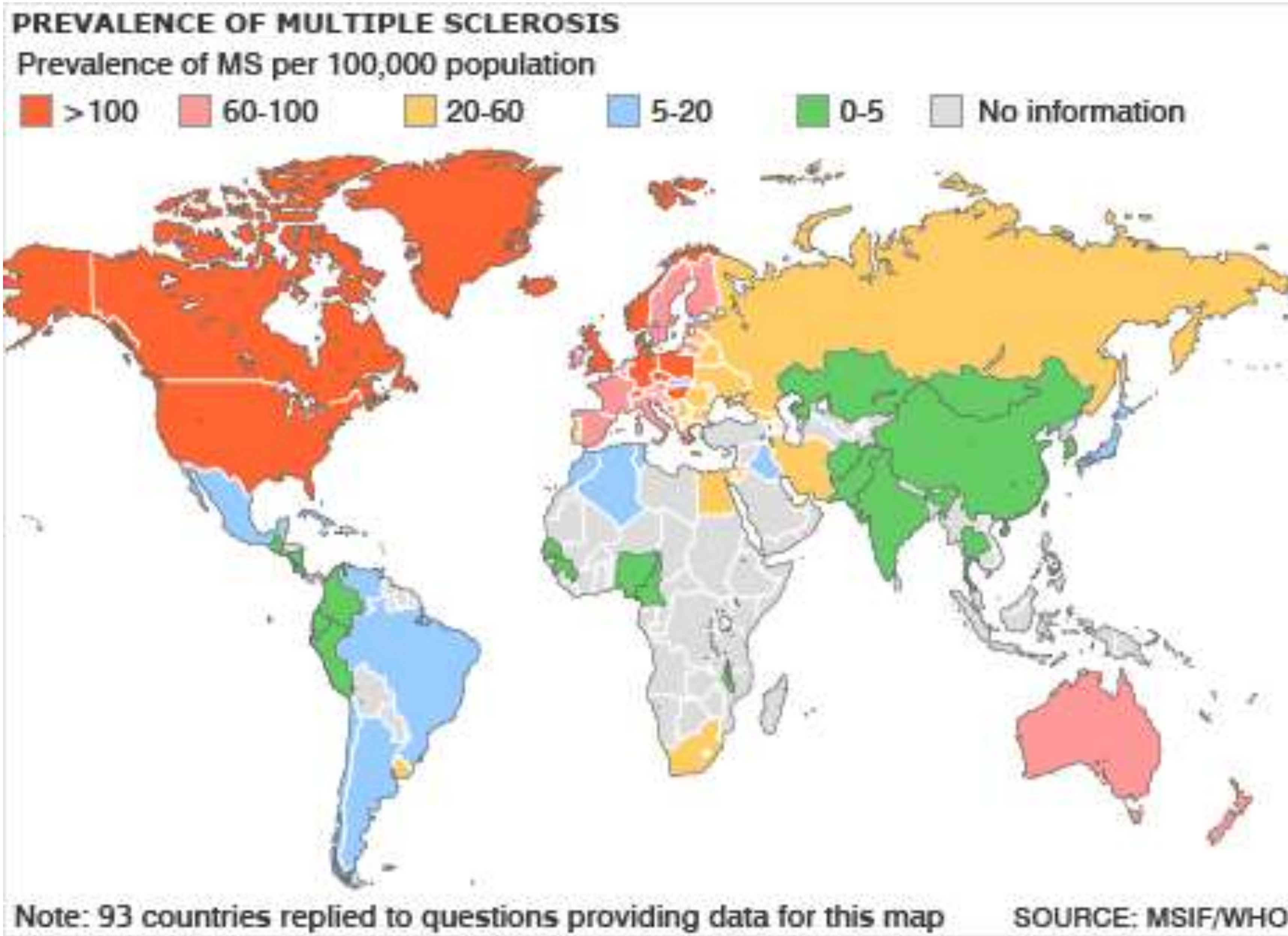
**Abnormal immunologic response → MS**

Gilden et al. *Lancet Neurol*. 2005;4:195, Noseworthy et al. *N Engl J Med*. 2000;343:938.

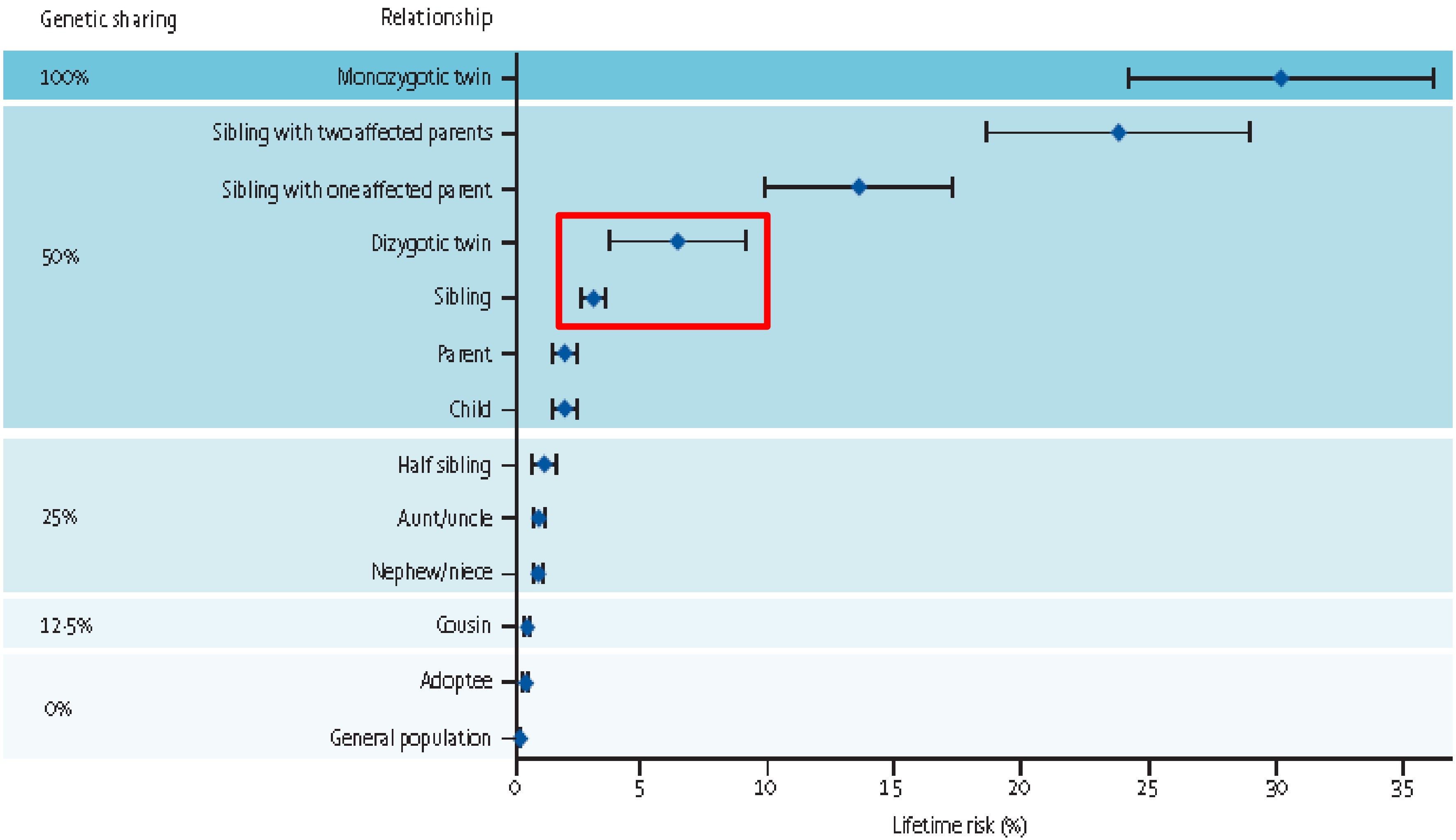
# Epidemiology

- Age of onset - $3^{rd}$ / $4^{th}$ decade (16 - 50 years)
- Prevalence – UK ~125/100,000  (latitude dependent)
- In an unpublished communication, Prof. Bohlega estimated the prevalence of MS in Saudis to be 40/100,000 in 2008.
- Life Span - slightly reduced (~ 10 years)
- Sex - F > M
- Race - Caucasians
  (uncommon in Chinese / ? Viking ancestral genes)
- Geography - Northern European Disease
- Familial clustering

Multiple sclerosis in the Arabian Gulf countries: a consensus statement Saeed Bohlega • Jihad Inshasi • Abdel Rahman Al Tahan • Abu Bakr Madani • Hussien Qahtani • Pet
Rieckmann. J Neurol (2013) 260:2959–2963


PREVALENCE OF MULTIPLE SCLEROSIS
Prevalence of MS per 100,000 population
>100
60-100
20-60
5-20
0-5
No information
Note: 93 countries replied to questions providing data for this map
SOURCE: MSIF/WHO

# Familial Risk




Compston & Coles, Lancet 2008.

The NEW ENGLAND JOURNAL of MEDICINE

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# Risk Alleles for Multiple Sclerosis Identified by a Genomewide Study

The International Multiple Sclerosis Genetics Consortium*

## HLA-DRB1

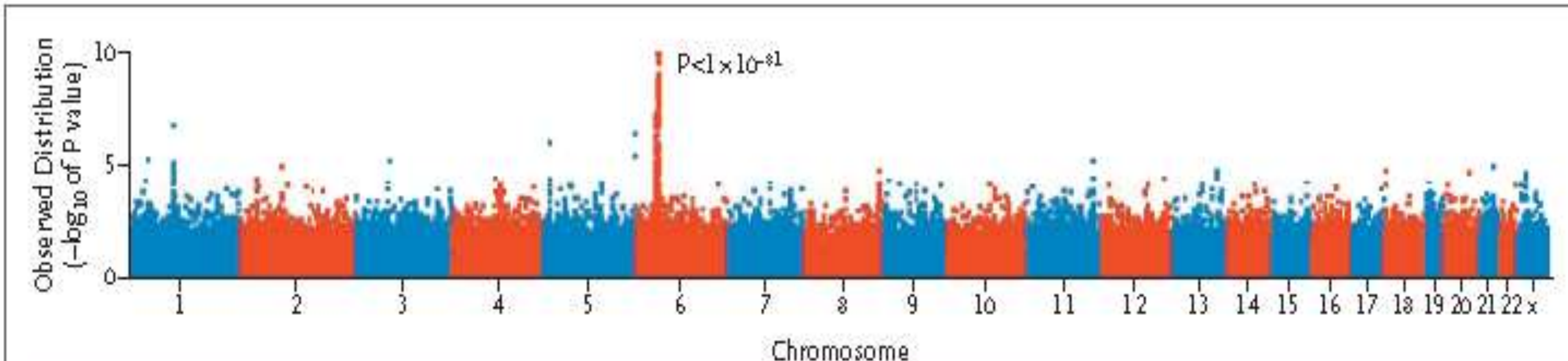



Figure 2. Overview of the Primary Genomewide Association Scan Involving 931 Family Trios.

P values (shown as $-\log_{10}$ values) for results of transmission disequilibrium testing are plotted across the genome. The classic HLA-DR risk locus on chromosome 6p21 stands out with strong statistical significance ($P<1\times10^{-81}$).

# Other Genetic contributions

HLA-DRB1 15.01 (polymorphism) OR = 3.1

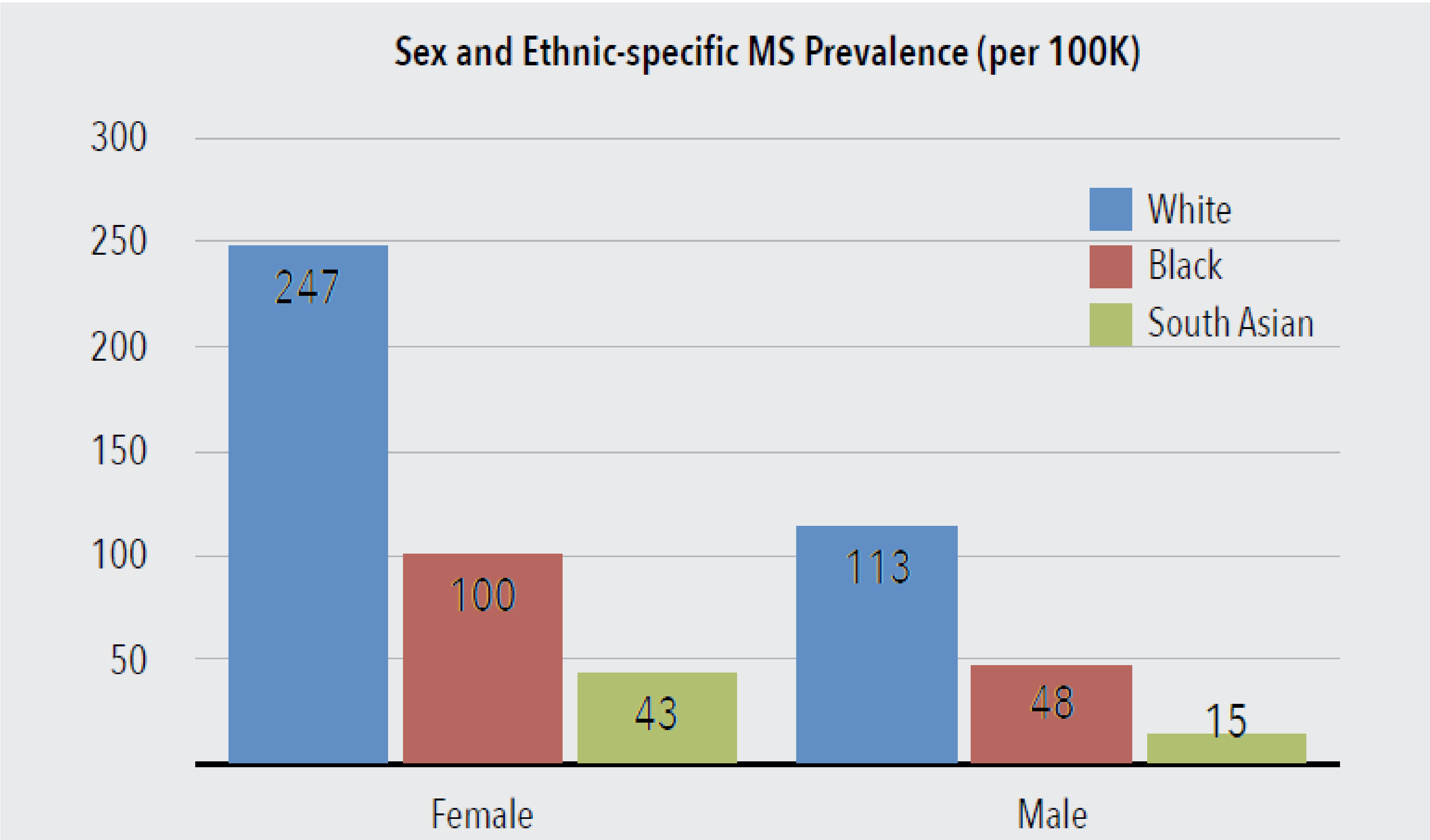
The MS Genome 2012

-> 100 non HLA-DRB1 polymorphism OR = 1.1 to 1.2

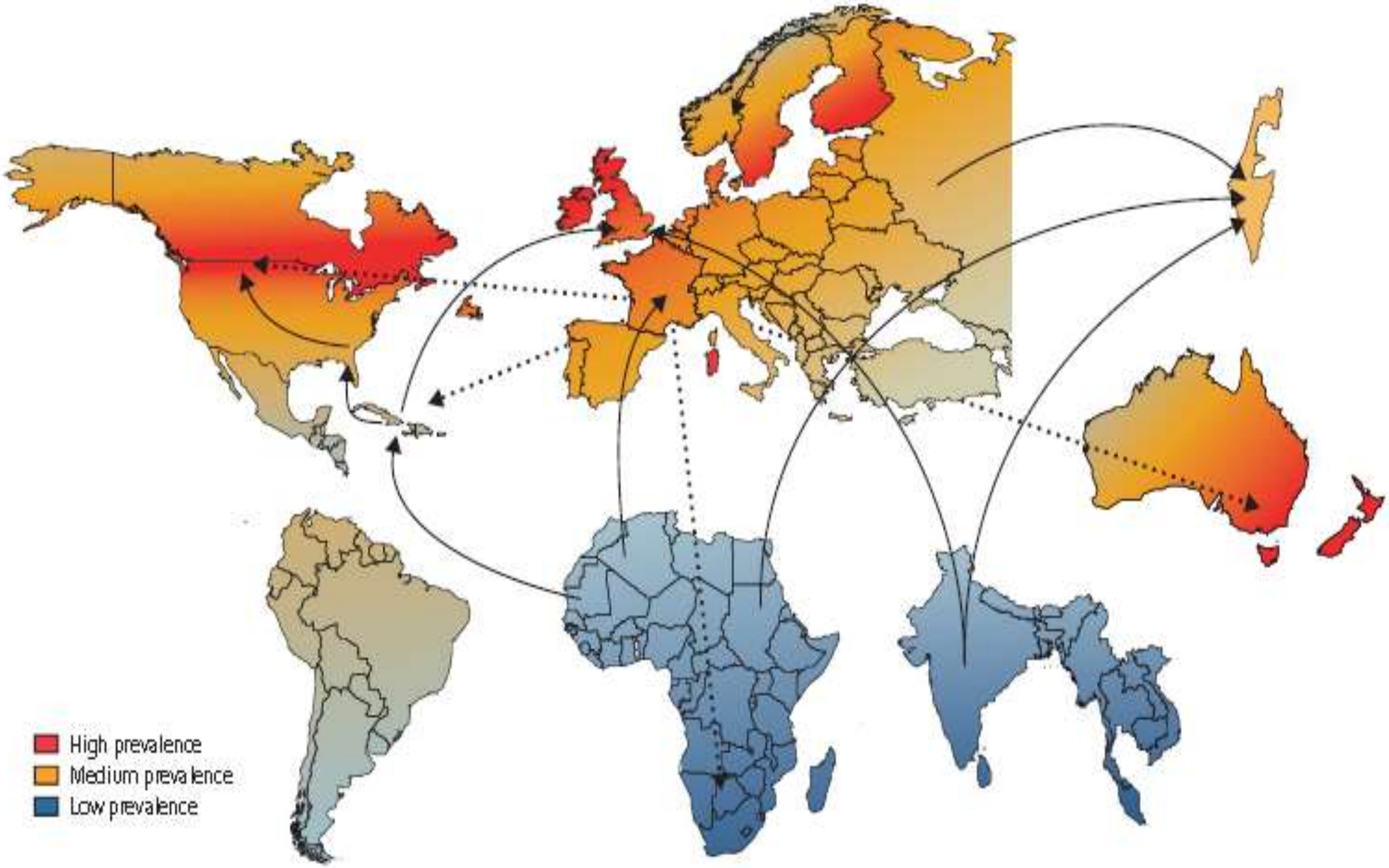
Odds ratio's small,  few %

Genes (15to20% influence risk)

# Barts MS ethic distribution



# Migration studies




Compston & Coles, Lancet 2008.

# Increasing female incidence

The changing demographic pattern of multiple sclerosis epidemiology

Nils Koch-Henriksen, Per Soelberg Sørensen

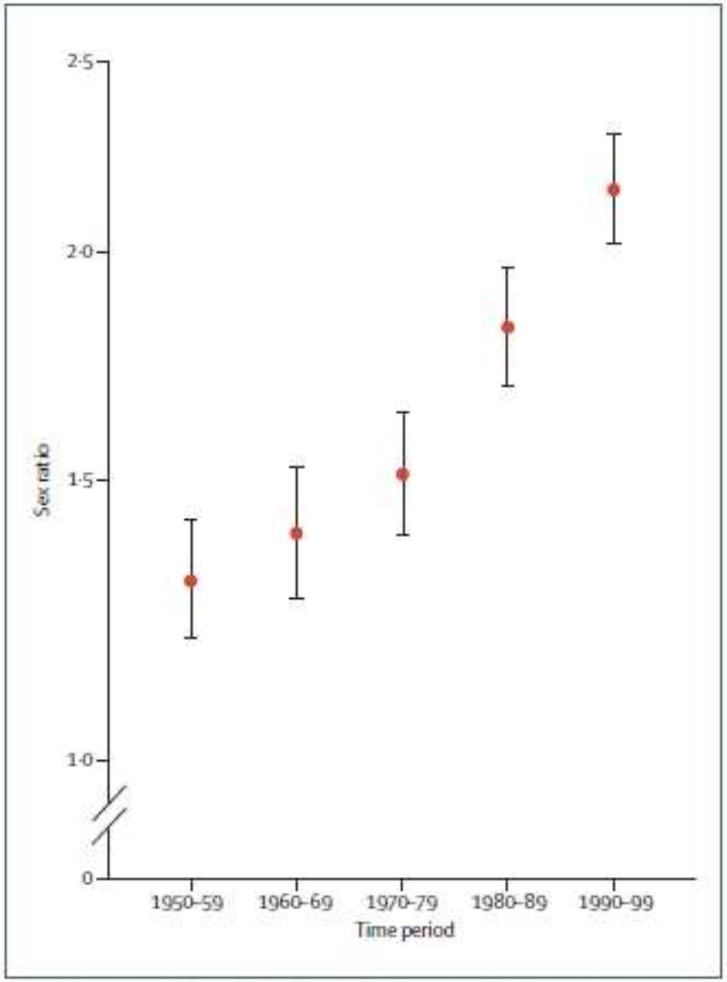
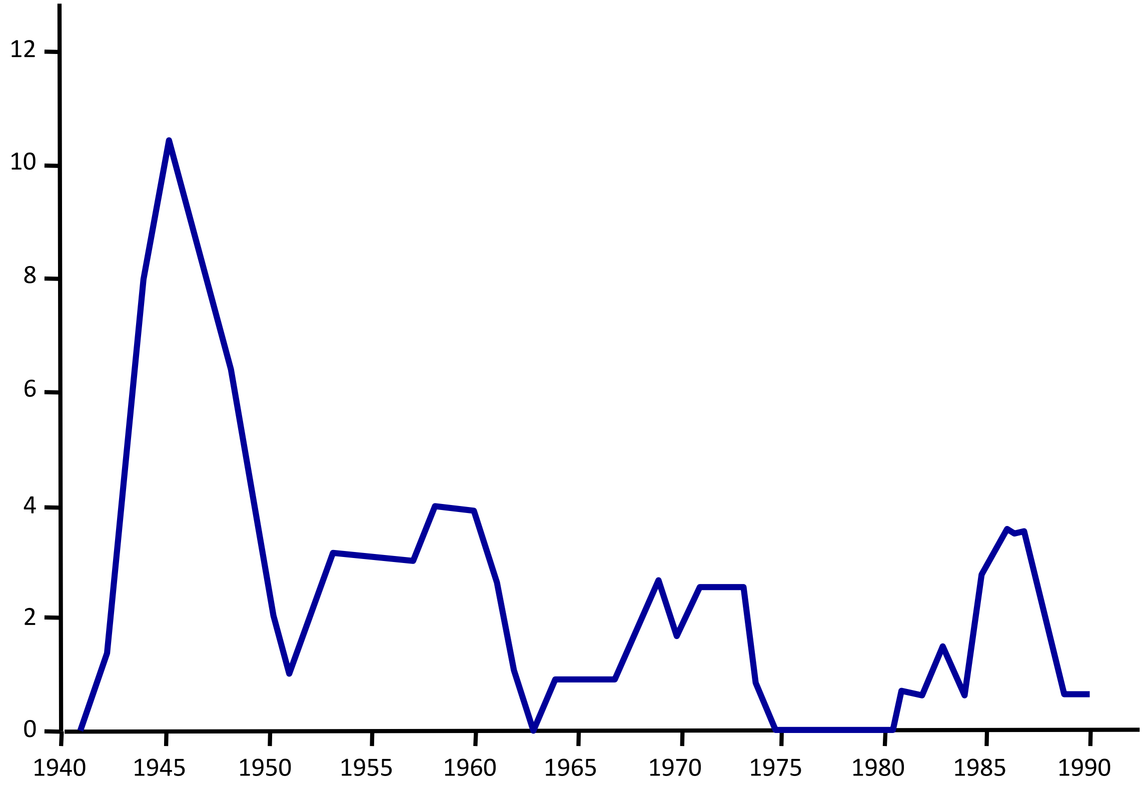



**Figure 8: Increase in female:male sex ratio in Denmark since 1950**
A marked increase in incidence of multiple sclerosis in women can be seen. Sex ratios (95% CIs) are shown on a logarithmic scale. Sex ratios are calculated as the ratio of life-time cumulative incidence proportions compiled on the basis of age-specific and sex-specific incidence rates in a given period. This method replaces standardisation and makes ratios at different times comparable.

# Epidemics or clusters of MS

The annual incidence of MS (per 100 000 inhabitants) in the Faroe Islands since 1940

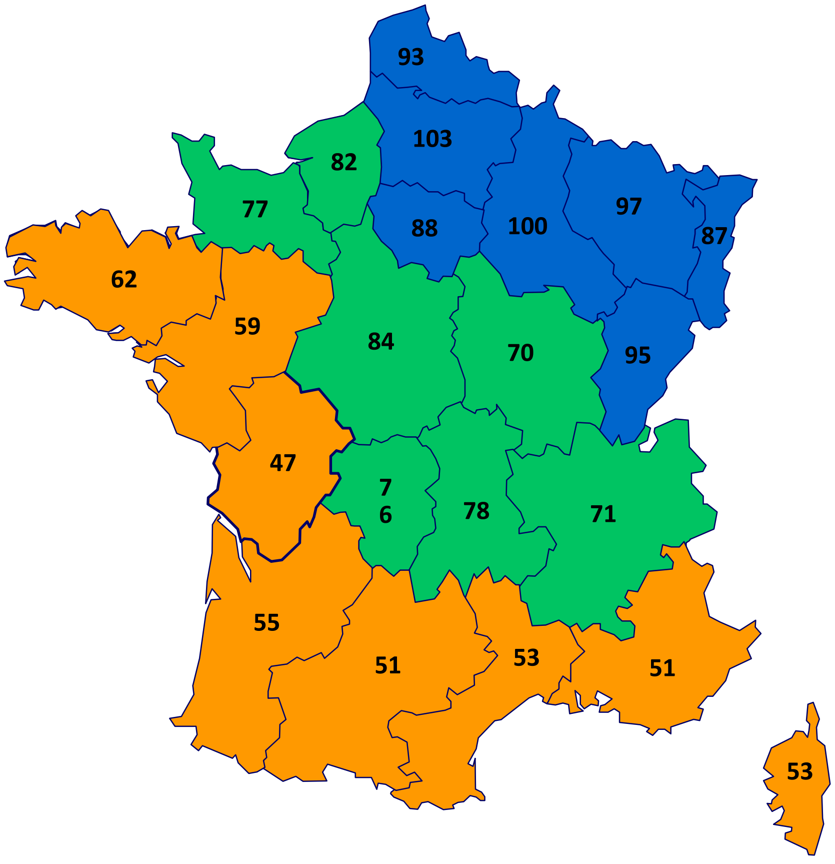



Kurtzke JF *et al. Acta Neurol Scand* 1993;**88:**161–173.

# Enviromental Risk Factors

- Sunlight/UVB
- Vitamin D insufficiency (x2  risks)
- Remote EBV (x 2-3 risk) 10% of paediatric MS EBV neg
- Smoking (increases 50 to 70%)
- Parity (decreased risk of no. of children) cumulative beneficial effect of pregnancy
- Obesity (lower Vit D)
- Salt – pro-inflammatory, not confirmed

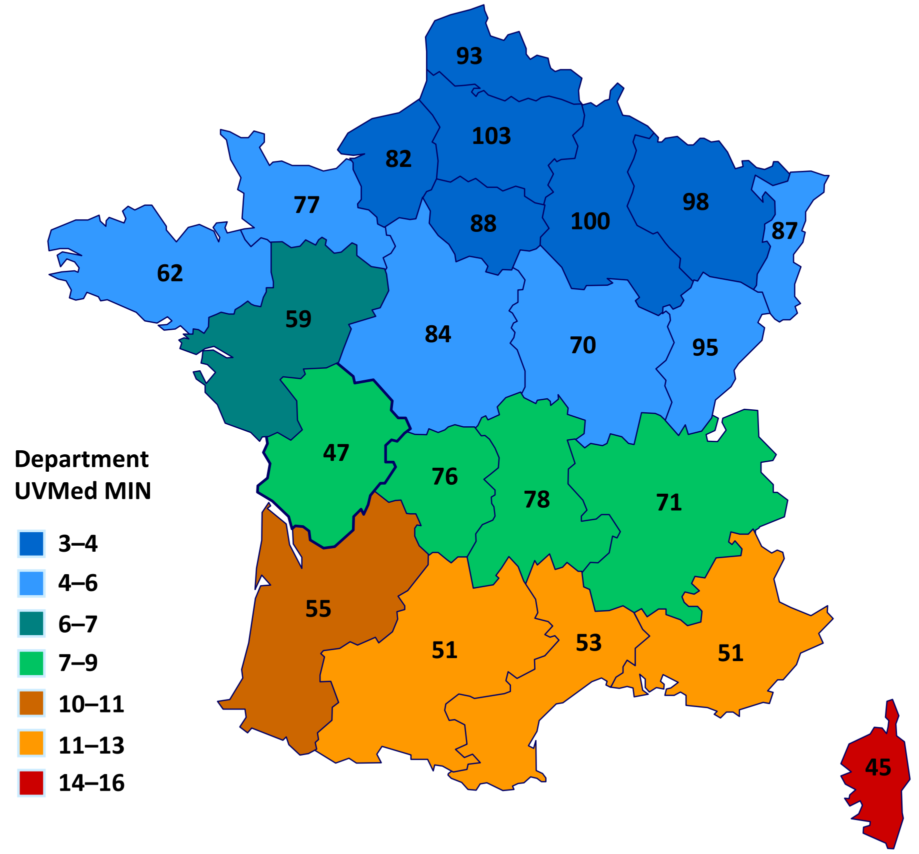
# Geographical distribution of MS: prevalence increases away from the equator




Vukusic S *et al. J Neurol Neurosurg Psychiat* 2007;**78:**707–709.

# Role of $vD_3$: UVB and MS prevalence

MS Prevalence by Department Against UVMED minimum




[1]Jablonski NG, Chaplin G. *J Hum Evol* 2000;**39:**57–106.
[2]Chaplin G. *Am J Phys Anthropol* 2004;125**:**292–302.

# Month of Birth

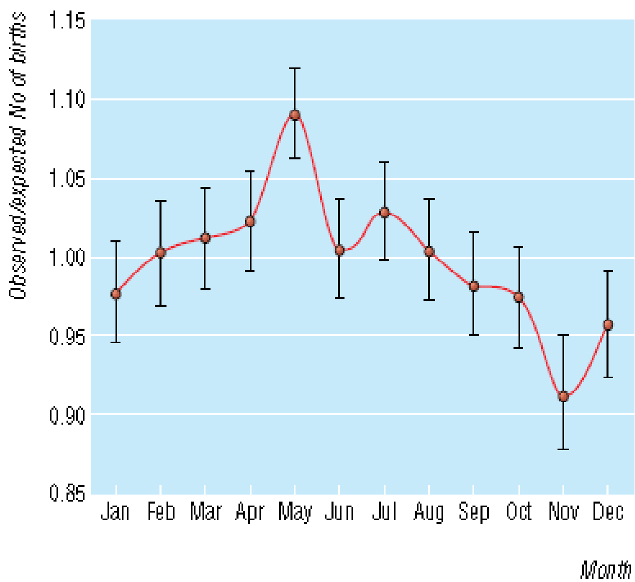



**Fig 1** Pooled analysis of observed/expected births in people with multiple sclerosis in Canadian, British, Danish, and Swedish studies (n=42 045) with 95% confidence intervals

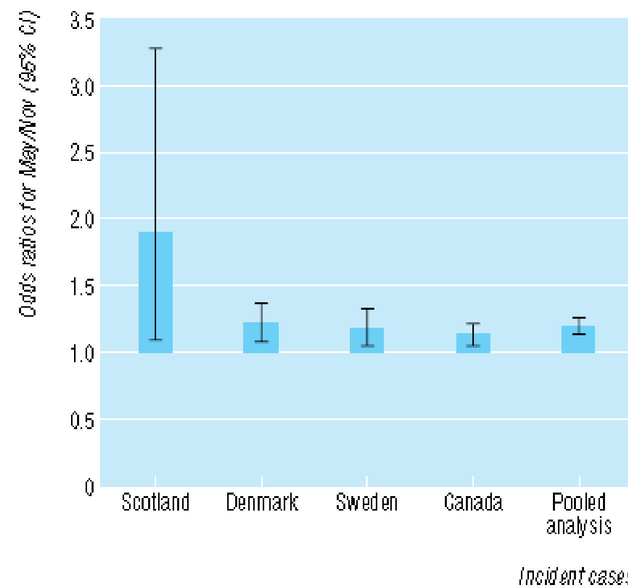
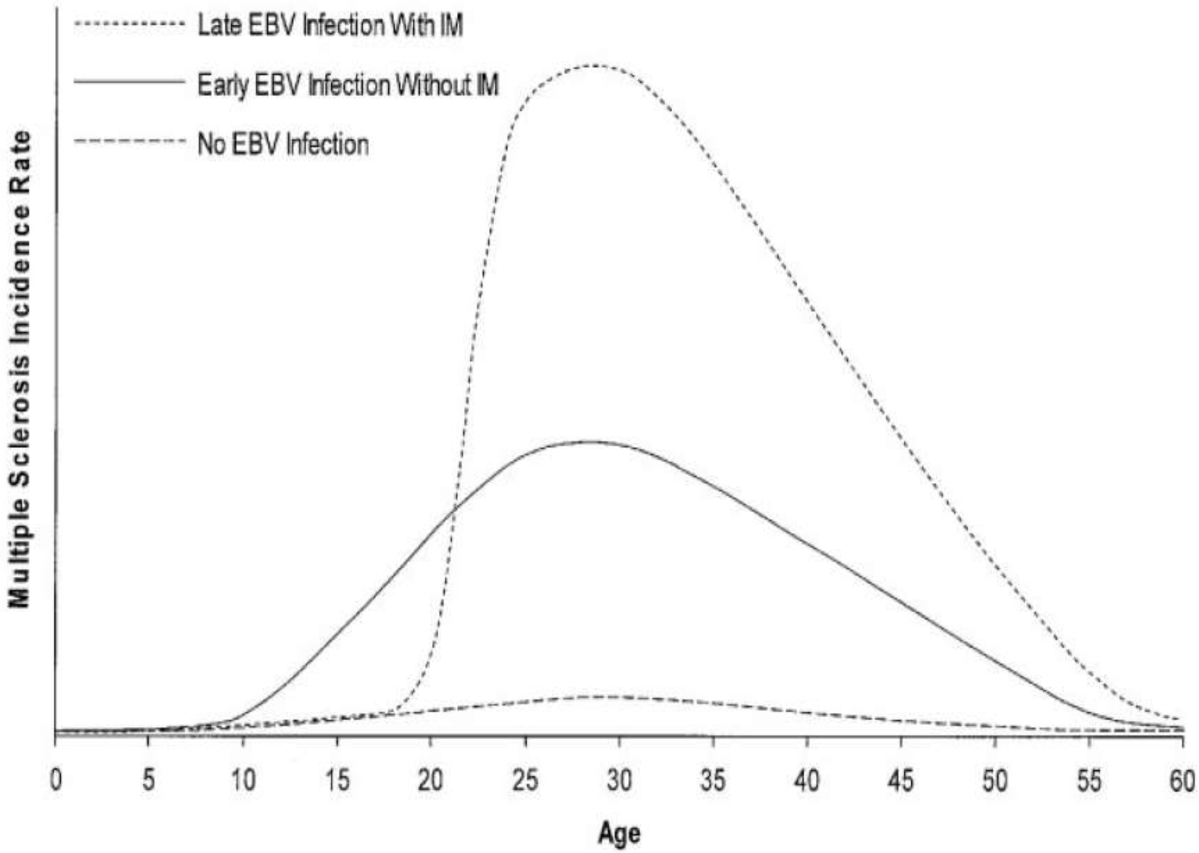



**Fig 2** Odds ratios for people with multiple sclerosis being born in May/November among incident cases in northern hemisphere countries

[1]Willer CJ *et al*. *BMJ* 2005;330:120–125.

# Symptomatic Epstein-Barr virus (EBV) infection and MS




Thacker et al. Ann Neurol 2006;59:499–503.

# Common viruses associated with lower pediatric multiple sclerosis risk.

**Table 5** Multivariate model including all available remote viral exposures (adjusted for age at blood draw, sex, race, and ethnicity)

| | OR to develop MS/CIS | 95% CI | p Value |
|---|---|---|---|
| Anti-EBNA-1-positive | 5.00 | 1.80–13.90 | 0.002 |
| Anti-CMV-positive | 0.30 | 0.11–0.77 | 0.01 |
| Anti-HSV-1-positive | 2.86 | 0.75–10.89 | 0.12 |
| *DRB1*1501/1503*-positive | 3.00 | 1.27–7.11 | 0.01 |

Abbreviations: CI = confidence interval; CIS = clinically isolated syndrome; CMV = cytomegalovirus; EBNA = Epstein-Barr nuclear antigen; HSV = herpes simplex virus; MS = multiple sclerosis; OR = odds ratio.

**Conclusions:** These findings suggest that some infections with common viruses may in fact lower MS susceptibility. If this is confirmed, the pathways for risk modification remain to be elucidated. E.Waubant et al ***Neurology***® **2011;76:1989–1995**

# ‘Hygiene Hypothesis’ of autoimmune disease

Ponsonby AL, van der Mei I, Dwyer T, et al. Exposure to infant siblings during early life and risk of multiple sclerosis. JAMA 2005;293:463–9

Pedrini MJF, Seewann A, Bennett KA, et al. Helicobacter pylori as a protective factor against multiple sclerosis risk in females. J Neurol Neurosurg Psychiatry 2015;86:603–7

# Gene-environment interactions

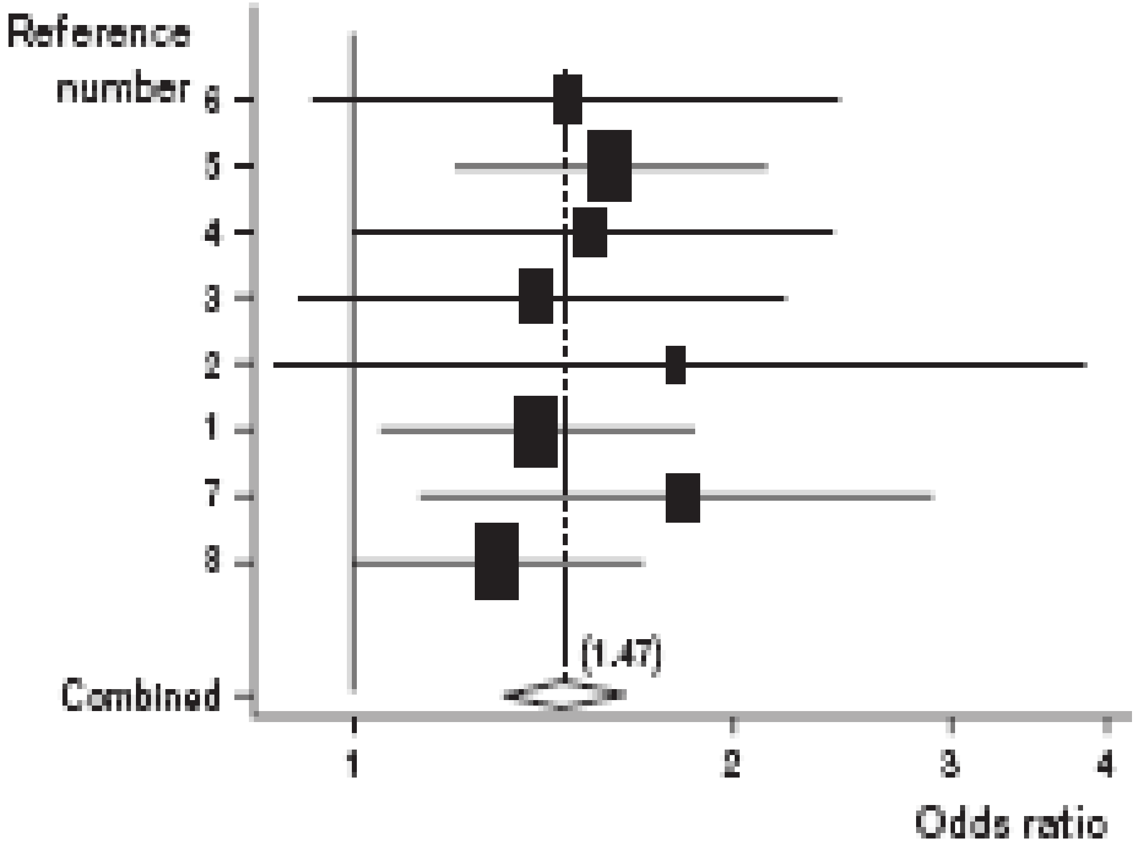
Epigenetic modification – changes in expression of DNA without alteration of DNA

Mechanisms:

- DNA methylation – decreases expression
- Micro RNA's – small sequence changes expression of RNA

Example : DRB1*1501 and adolescent obesity (OR 16)

# Smoking is a risk factor for multiple sclerosis



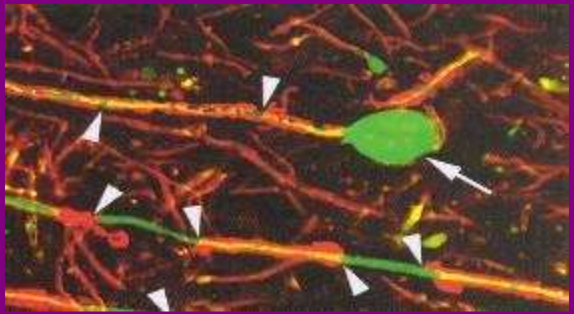

Hawkes CH, Mult Scler. 2007 Jun;13(5):610-5.

# Multiple Sclerosis: 3 Components

**Inflammation**

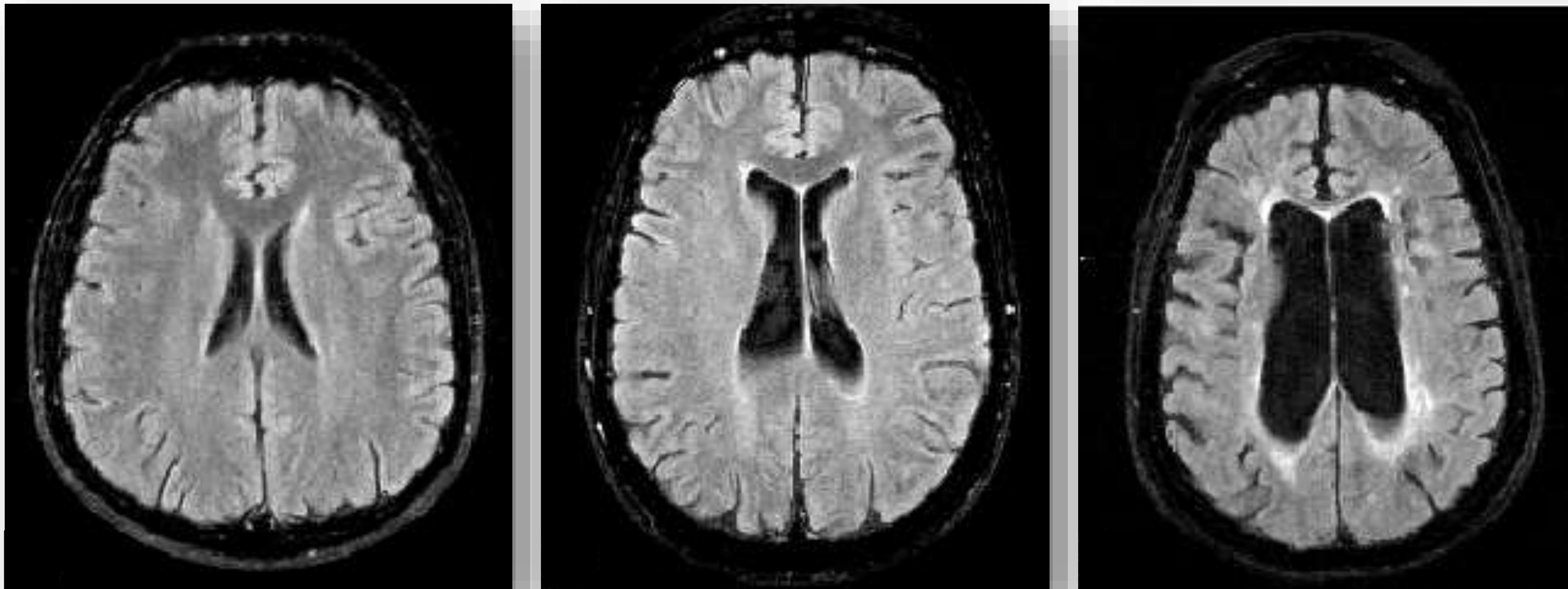
**Demyelination**

**Axonal Loss**



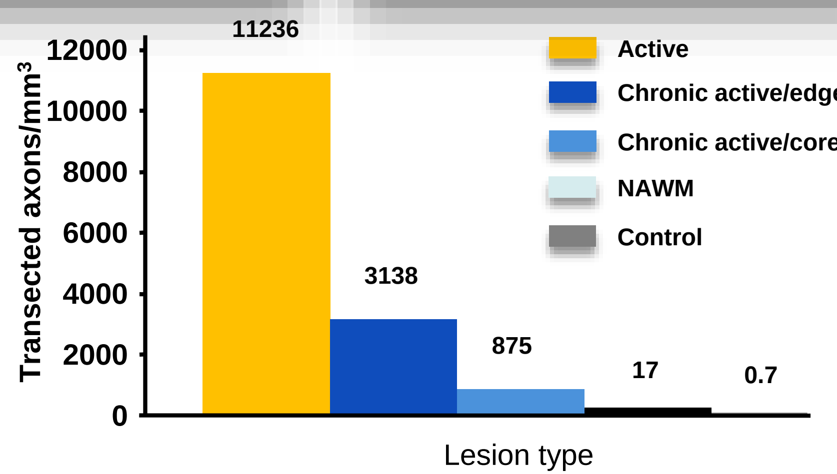
Trapp et al. *Curr Opin Neurol*. 1999;12:295; Trapp et al. *J Neuroimmunol*. 1999;98:49; Trapp et al. *Neuroscientist*. 1999;5:48.

# Inflammation causes Axonal Damage and Atrophy



**Average reduction in brain volume[1]**

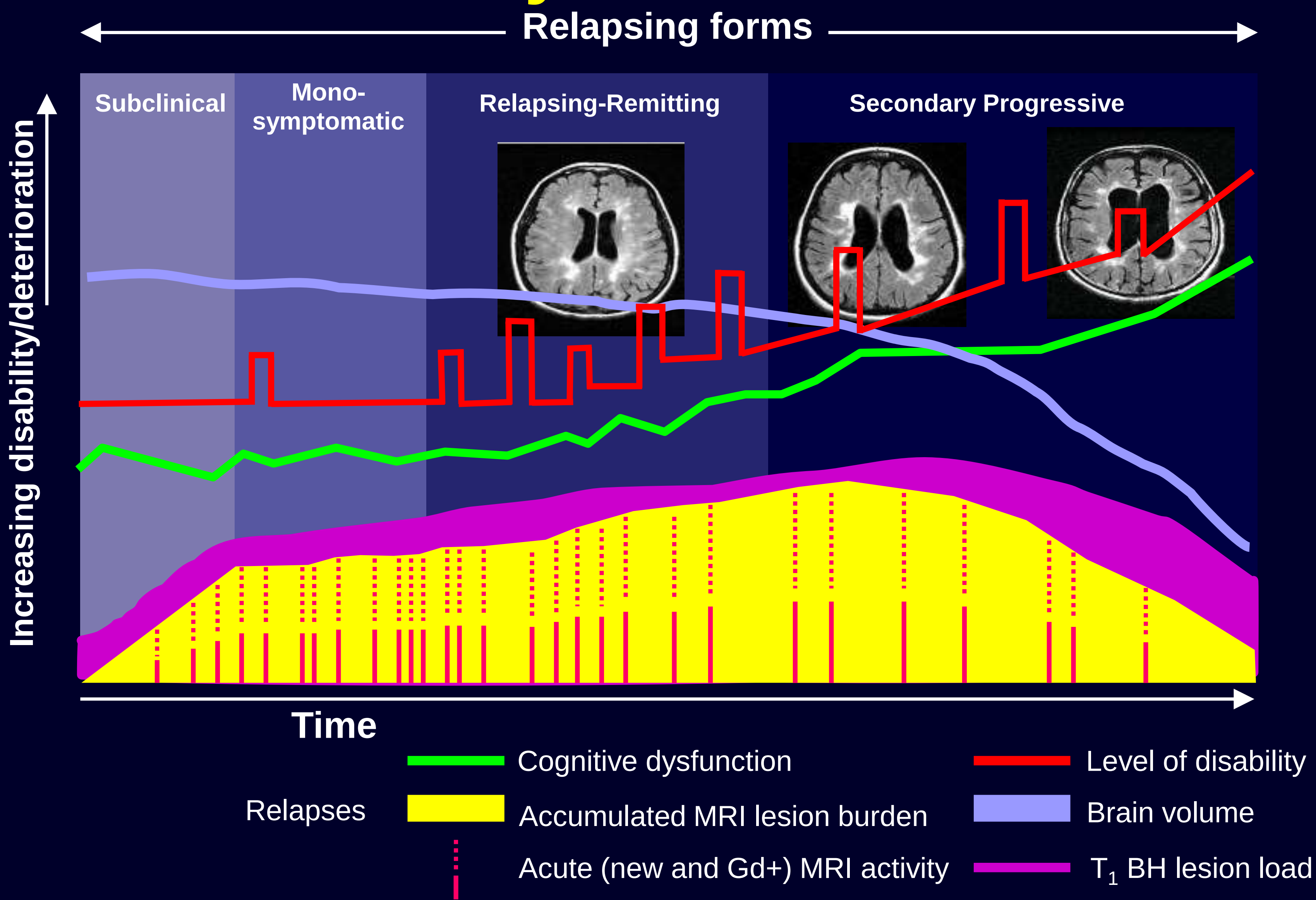
| | |
|---|---|
| MS: | 0.6 – 1.0% per year |
| Healthy: | 0.1 – 0.3% per year |




NAWM=normal-appearing white matter


*1. Miller DH et al. Brain 2002;125:1676-1695;*
*2. Trapp BD et al. New Engl J Med. 1998;338:278-285.*

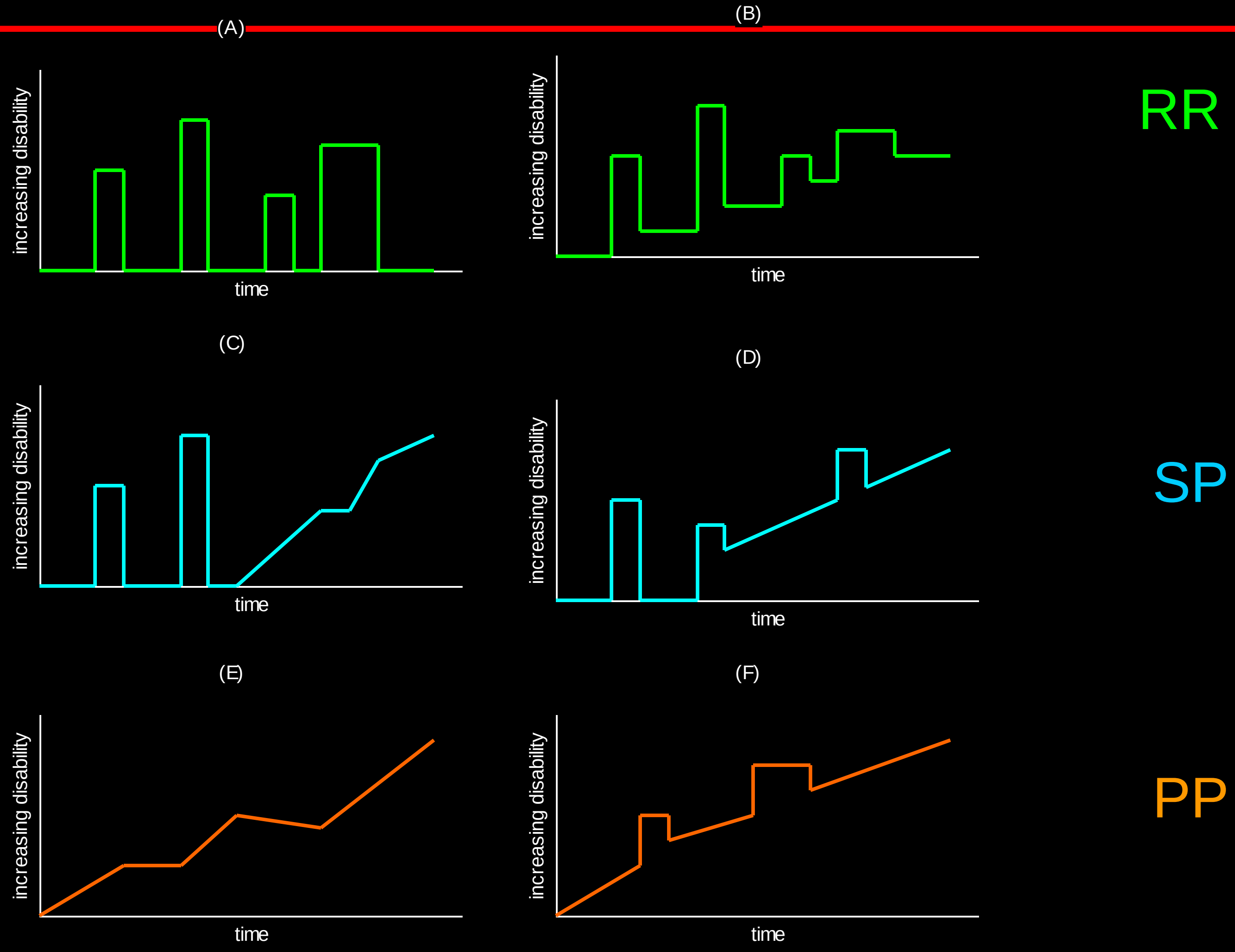
# Natural History of MS



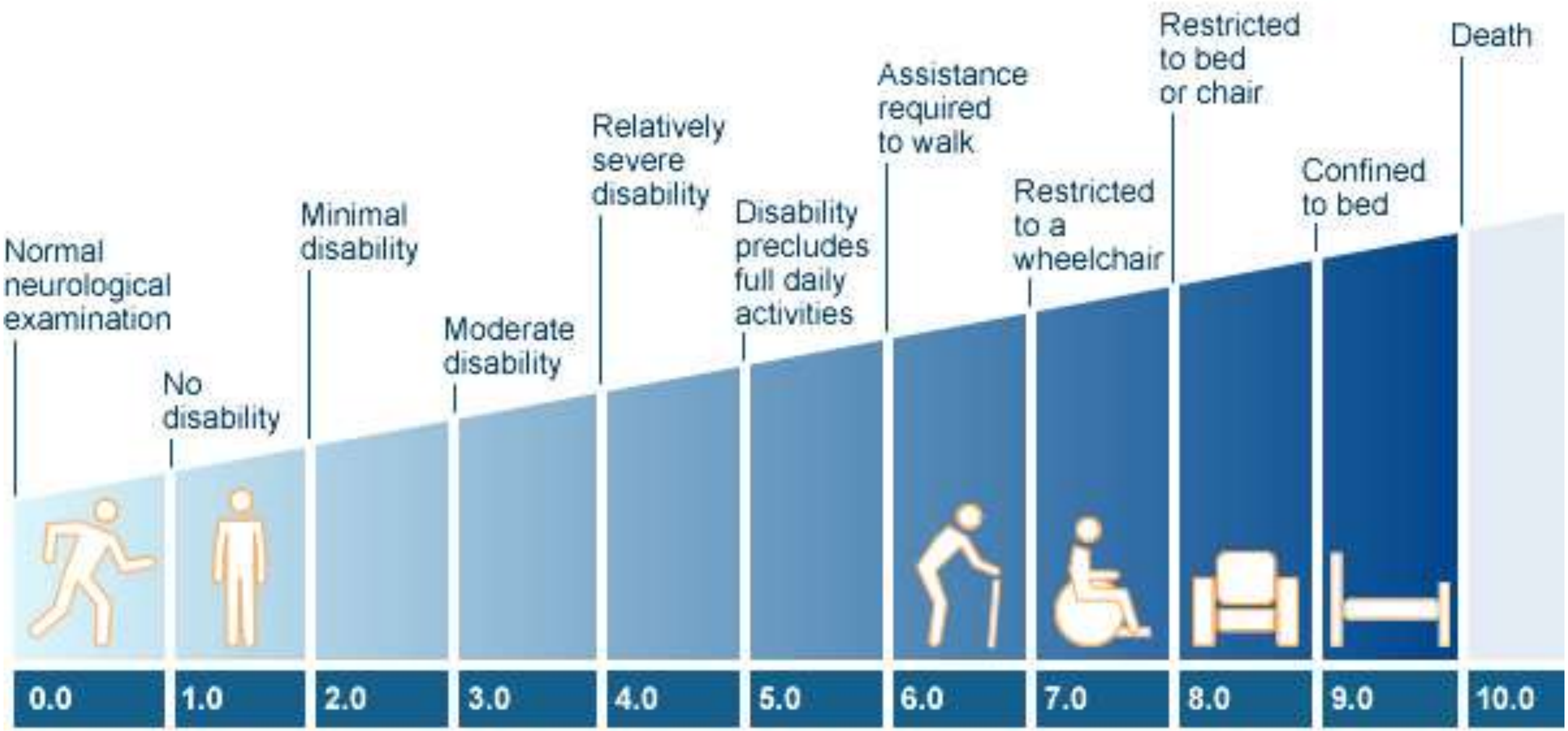

BH = black hole; Gd = gadolinium; MRI = magnetic resonance imaging.

Noseworthy et al. *N Engl J Med.* 2000;343:938; Weinshenker et al. *Brain.* 1989;112:133; Trapp et al. *Curr Opin Neurol.* 1999;12:295.

# Clinical courses of MS



# MS Expanded Disability Status Scale - EDSS

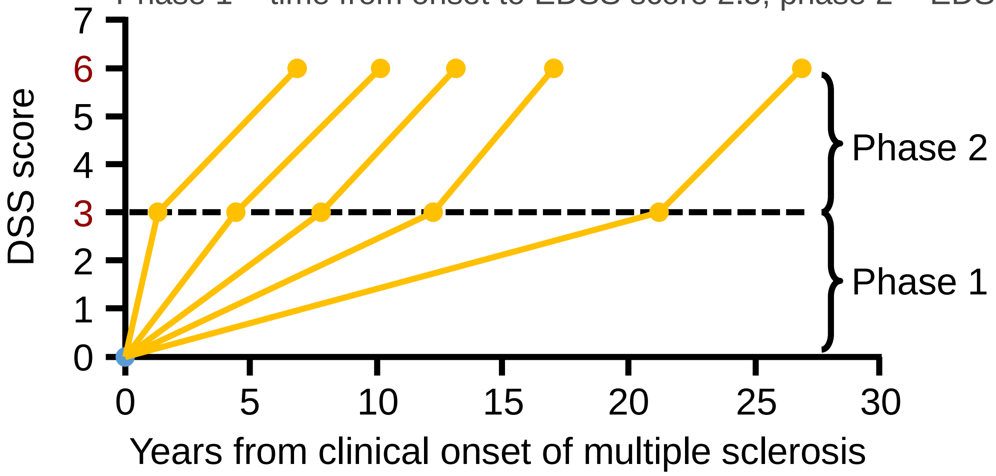


# EDSS score <3.0 as a treatment goal: suggested approach illustrates limitations of EDSS

**Traditionally, EDSS 4.0 has been seen as a milestone for disability progression, but it has been suggested that a new treatment goal may be to keep patients at an EDSS score of 3.0 or less[1]**

**This arises from the school of thought that disability progression is a 2-stage process:[1]**

Phase 1 = time from onset to EDSS score 2.5; phase 2 = EDSS score 3.0−6.0



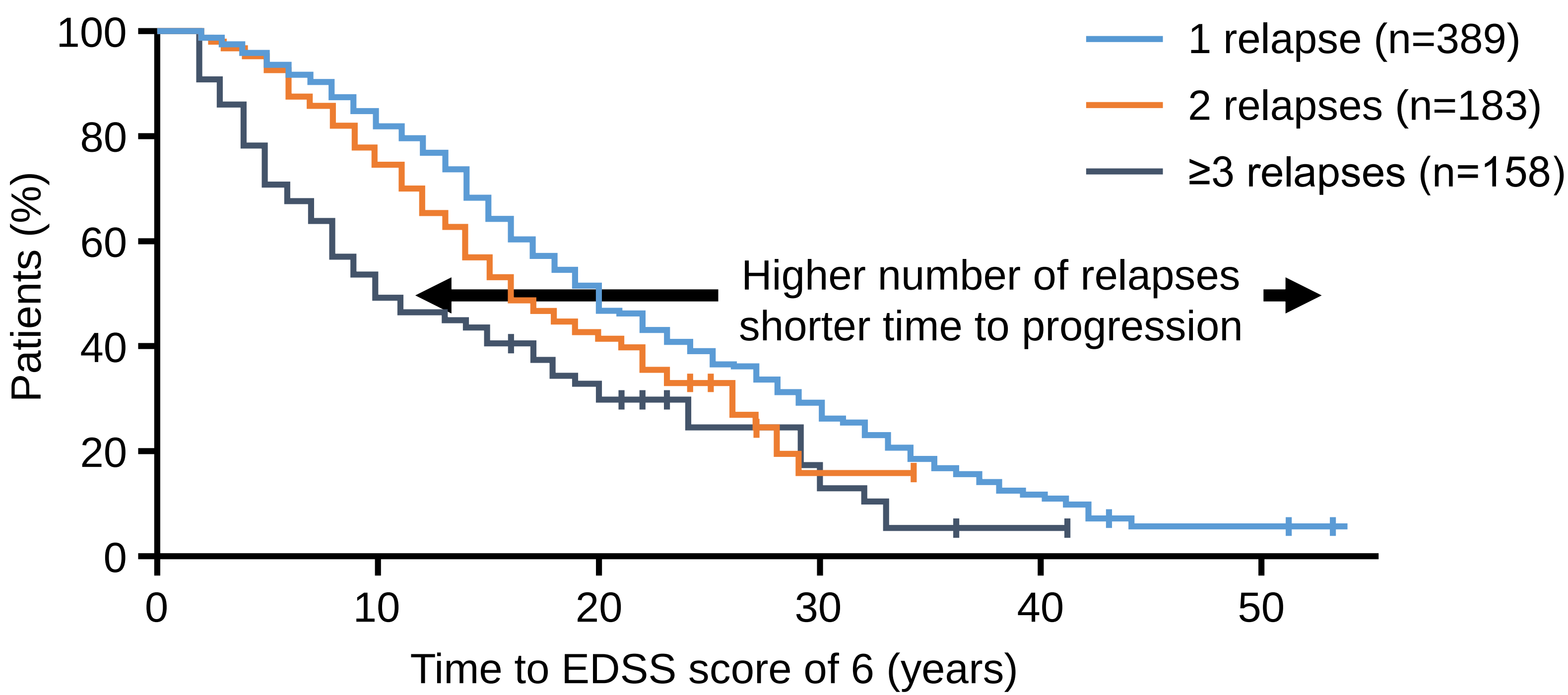

Disability progression in Phase 2 in five groups of patients (N=718) defined by duration of Phase 1[1]

- However, the 2-stage distinction may be artifactual, caused by the nature of the EDSS itself
  – Over-reliance on motor function assessment prior to EDSS 7.0[2]

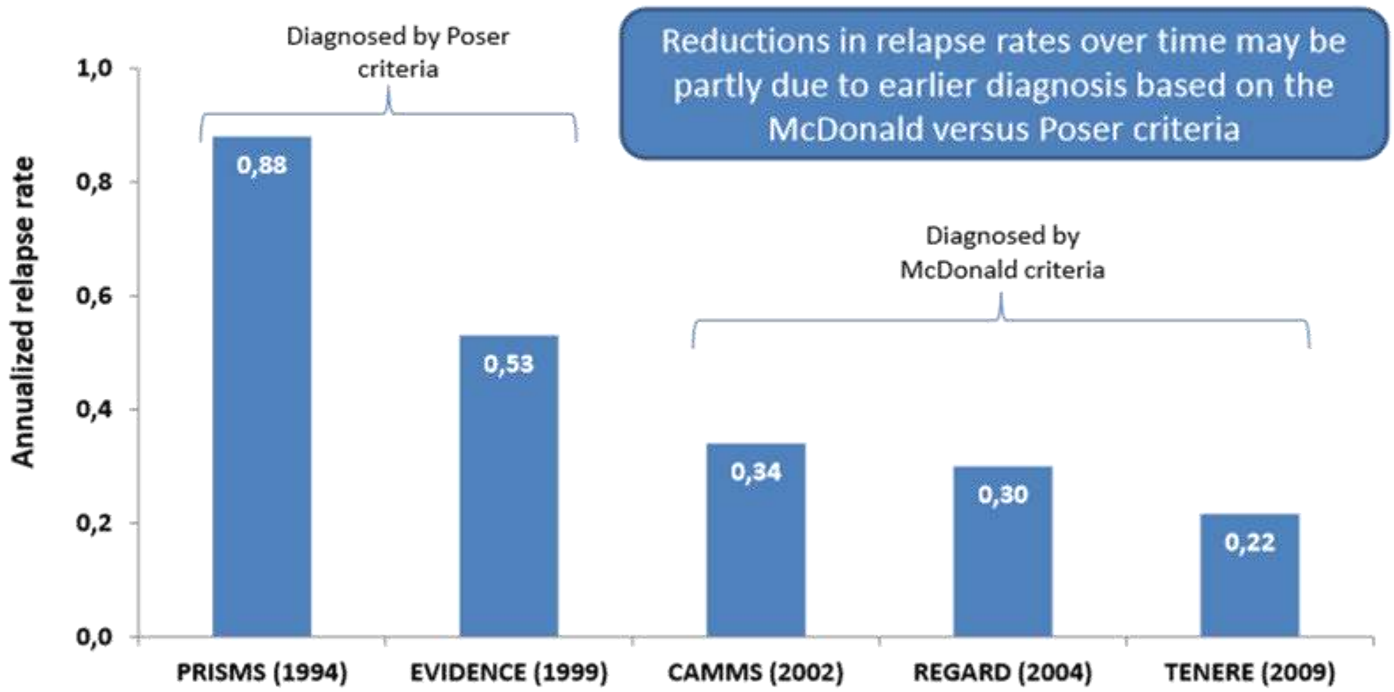
# Early relapse activity is associated with long-term disability progression in natural history cohorts[1–5]

**Natural history data over a mean follow up of nearly 3 decades show that number of relapses in the first 2 years after diagnosis correlates with time to disability progression[3]**

Stratification based on the number of relapses in first 2 years[4]



# Rate of relapse has reduced over time with high dose of INF beta in RRMS studies




Annualized relapse rates in the treatment arms of Rebif® 44 µg subcutaneously three times weekly, across RRMS studies (with year that recruitment started)

Uitdehaag BMJ et al. Curr Med Res Opin 2011;27:1529-37

# Prognosis for MS

**‘after 15 years 50 % of MS patients will require an aid to walk (EDSS 6.0)’**

Weinshenker et al. *Brain.* 1989;112:133, Runmarker and Anderson. *Brain.* 1993;116:117

**Median EDSS after 14.1 years 3.25**

Brex et al. *N Engl J Med*. 2002;346:158.

**Median time to EDSS 6 in European population of 2,495 patients 27.3 years**

Debouverie et al Neurology 2007;68:29

# Prognosis for MS

## Clinical predictors of disease course

**Multifocal presentations**

**Interval between 1st and 2nd relapse**

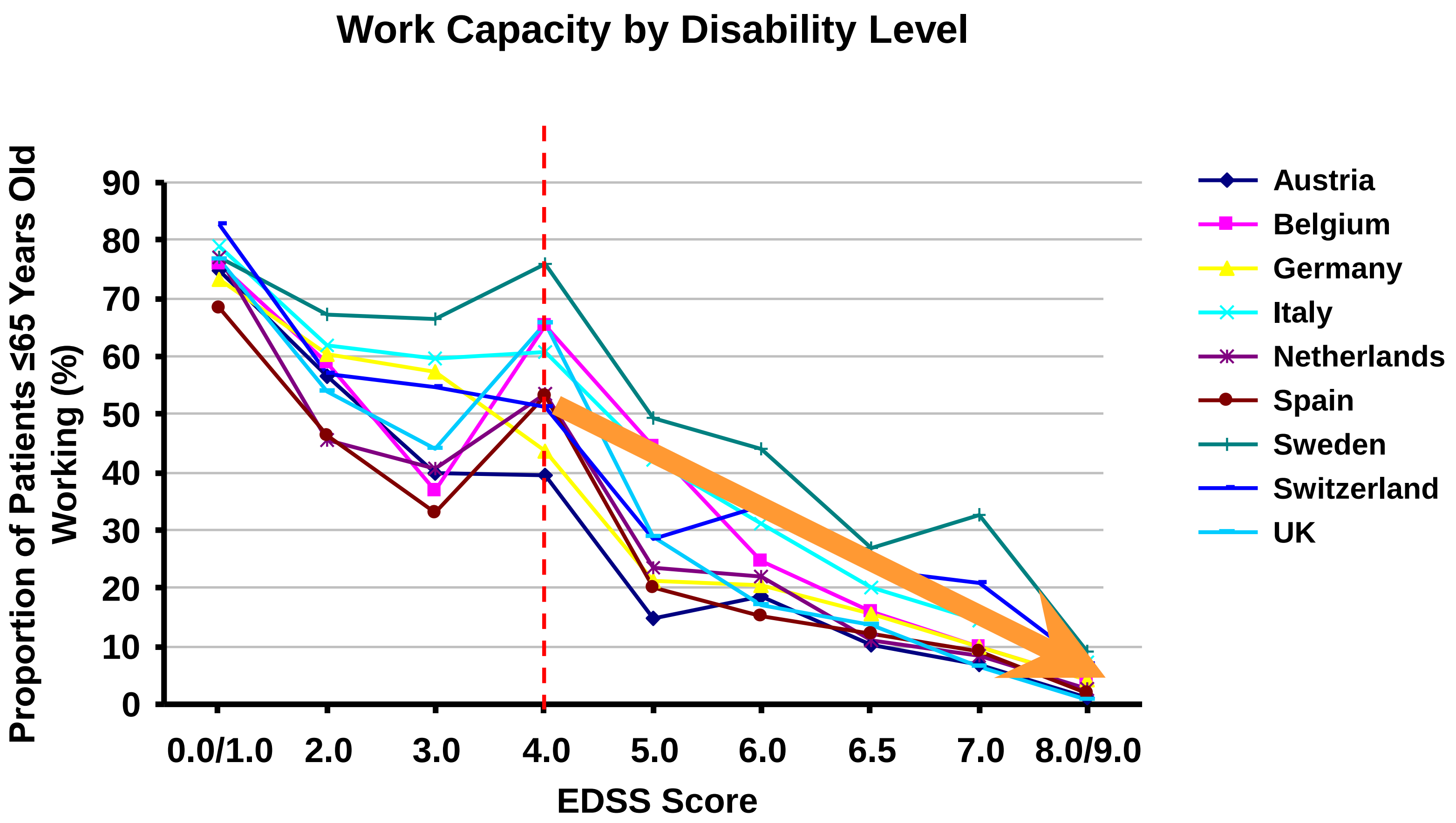
**Relapse frequency**

**Relapse recovery**

**Progression of disability in the first five years of disease**

Weinshenker et al. *Brain.* 1989;112:133, Runmarker and Anderson. *Brain.* 1993;116:117

# Employment Change With EDSS Score




The proportion of patients employed or on long-term sick leave is calculated as a percentage of patients aged 65 or younger.
The reduced work capacity particularly at EDSS 2.0–3.0 compared with EDSS 4.0 is explained by an increased frequency of relapses in this group and hence a concentration of sick leaves.

*Kobelt G. J Neurol Neurosurg Psychiatry. 2006;77:918-926.*

# Poser Diagnostic Criteria (1983)

**Clinical Definite MS – clinically definite MS by Poser Criteria**

**2 attacks and one paraclinical evidence (MRI, evoked potentials)**

**Laboratory-supported definite MS**

**I attack and one paraclinical evidence and oligoclonal bands**

**Clinically probable MS**

**Laboratory supported probable MS**

# Patients Presenting With CIS: The Queens Square Studies

n=109

5-year follow-up

n=89

10-year follow-up

n=81

14-year follow-up

n=71

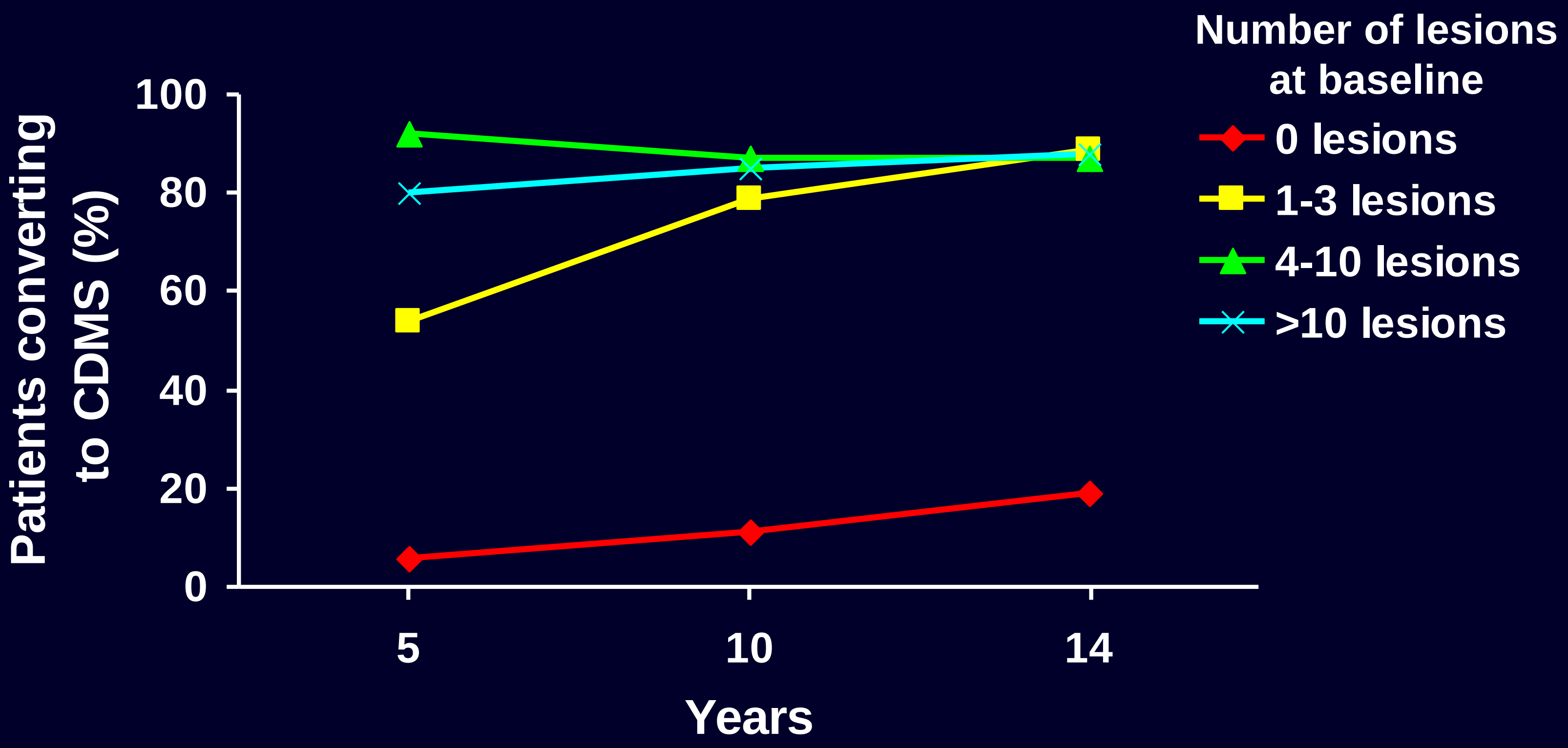
20-year follow-up

n=80

109 patients with CIS between 1984 and 1987 were enrolled and followed with brain MRI scans for a mean of 19.5 years

Miller et al. *J Neurol Neurosurg Psychiatry*. 1988;51:1569. Morrissey et al. *Brain*. 1993;116:135. O'Riordan et al. *Brain*. 1998;121:495. Brex et al. *N Engl J Med*. 2002;346:158;  Fisniku et al. *ENS*. 2006.

# Prognostic Significance of MRI in CIS: Rate of Conversion to CDMS




The data presented for years 5, 10, and 14 were obtained from different publications based on the same longitudinal study.


Morrissey et al. *Brain*. 1993;116:135; O'Riordan et al. *Brain*. 1998;121:495; Brex et al. *N Engl J Med*. 2002;346:158.

# 20-Year Follow-up in Patients With CIS

| | baseline | 5.3 yrs | 14.1yrs | 20yrs |
|---|---|---|---|---|
| No. of patients | 109 | 89 | 71 | 87 |
| CDMS (%) | 0 | 38 (43) | 48(68) | 55(63) |
| CDMS abnormal MRI at CIS (%) | 0 | 37 (65) | 44 (88) | 49 (86) |
| CDMS normal MRI at CIS (%) | 0 | 1 (3) | 4 (19) | 6 (20) |
| EDSS CDMS cases (median) | | 2.5 | 3.25 | 40% < 3<br>40 % 7.5 |

Brex et al. *N Engl J Med*. 2002;346:158. Fisniku et al ENS 2006

# Clinically Isolated Syndrome

**Definition -** a first neurological event suggestive of Multiple Sclerosis lasting for at least 24 hours and with symptoms and signs indicating a single or multiple lesions

e.g. – optic neuritis, spinal cord syndrome, brainstem lesion – INO

| | |
|---|---|
| Sensory Symptoms – | 34 % |
| Weakness | 22 % |
| Ataxia | 11 % |
| Visual symptoms | 13 % |
| Diplopia | 8 % |
| Bladder symptoms | 1 % |

# Diagnostic Criteria

## Diagnostic Criteria for Multiple Sclerosis: 2010 Revisions to the McDonald Criteria

Chris H. Polman, MD, PhD,[1] Stephen C. Reingold, PhD,[2] Brenda Banwell, MD,[3] Michel Clanet, MD,[4] Jeffrey A. Cohen, MD,[5] Massimo Filippi, MD,[6] Kazuo Fujihara, MD,[7] Eva Havrdova, MD, PhD,[8] Michael Hutchinson, MD,[9] Ludwig Kappos, MD,[10] Fred D. Lublin, MD,[11] Xavier Montalban, MD,[12] Paul O'Connor, MD,[13] Magnhild Sandberg-Wollheim, MD, PhD,[14] Alan J. Thompson, MD,[15] Emmanuelle Waubant, MD, PhD,[16] Brian Weinshenker, MD,[17] and Jerry S. Wolinsky, MD[18]

New evidence and consensus has led to further revision of the McDonald Criteria for diagnosis of multiple sclerosis. The use of imaging for demonstration of dissemination of central nervous system lesions in space and time has been simplified, and in some circumstances dissemination in space and time can be established by a single scan. These revisions simplify the Criteria, preserve their diagnostic sensitivity and specificity, address their applicability across populations, and may allow earlier diagnosis and more uniform and widespread use.

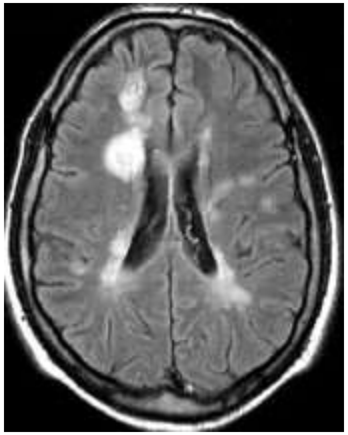
ANN NEUROL 2011;69:292–302

# DIS
# 2010 criteria

spinal

juxta

infra



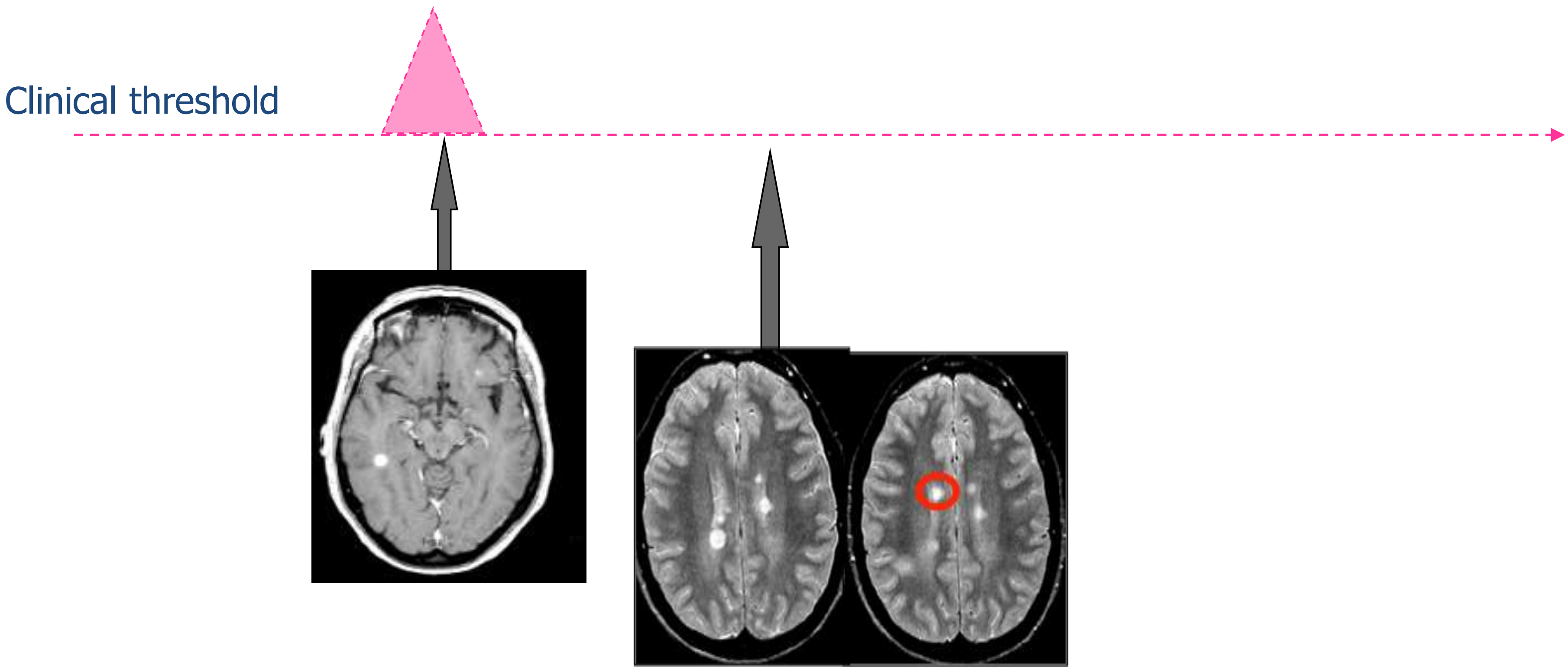
PV

≥ 1 asymptomatic T2 lesion in ≥ 2 topographies

# DIT
# 2010 criteria

A new T2 on follow-up MRI, with reference to a baseline scan, irrespective of the timing of the baseline MRI

Simultaneous presence of asymptomatic gadolinium-enhancing and nonenhancing lesions at any time



# New McDonald Criteria 2010

**Dissemination in Space (DIS)**

**$\geq$ 1 T2 lesion in at least 2 out of four MS-typical regions of CNS**
*periventricular,*
*juxtacortical,*
*infratentorial,*
*or spinal cord*
**does not include 9 T2 lesion cut-off**

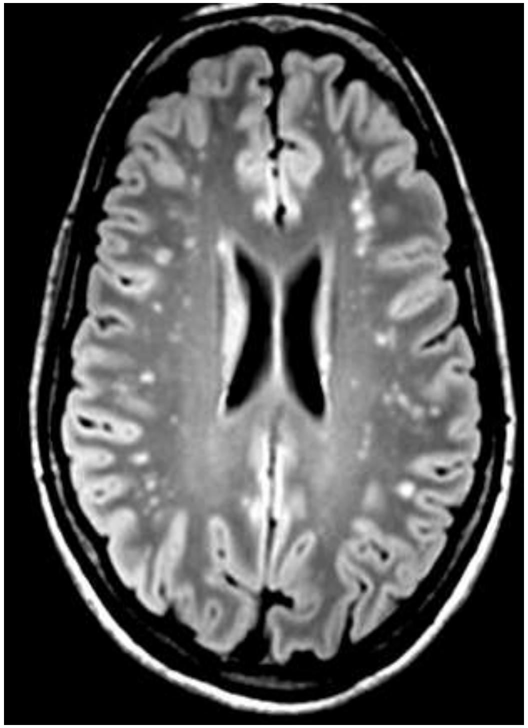
**Dissemination in time (DIT)**

**Simultaneous presence of asymptomatic Gd-enhancing AND non-enhacing lesions** at any time
*Or:*
**A *new* T2 and/or Gd+ lesion(s) on FU MRI** irrespective of timing of baseline scan

*Polman et al. Ann Neurol 2011;69:292–302*

# MRI focal white matter lesions

Incidental focal white matter lesions are far more common than **<u>multiple sclerosis</u>**



Multiple sclerosis:
Prevalence **<u>0.1/0.2%</u>** (20-40 years)

MRI focal white matter lesions (incidental):
Prevalence **<u>5-10%</u>** (20-40 years)

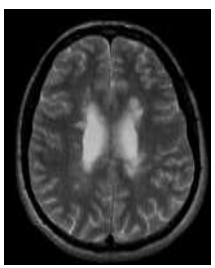
*Charil et al. Lancet Neurol 2006*

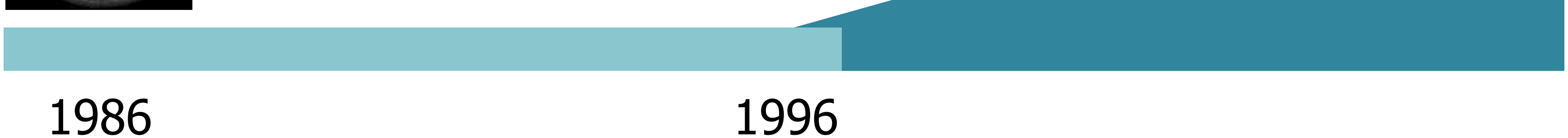
# Preclinical PPMS

*Clinical presentation of primary progressive multiple sclerosis 10 years after the incidental finding of typical magnetic resonance imaging brain lesions*

The subclinical stage of primary progressive multiple sclerosis may last 10 years

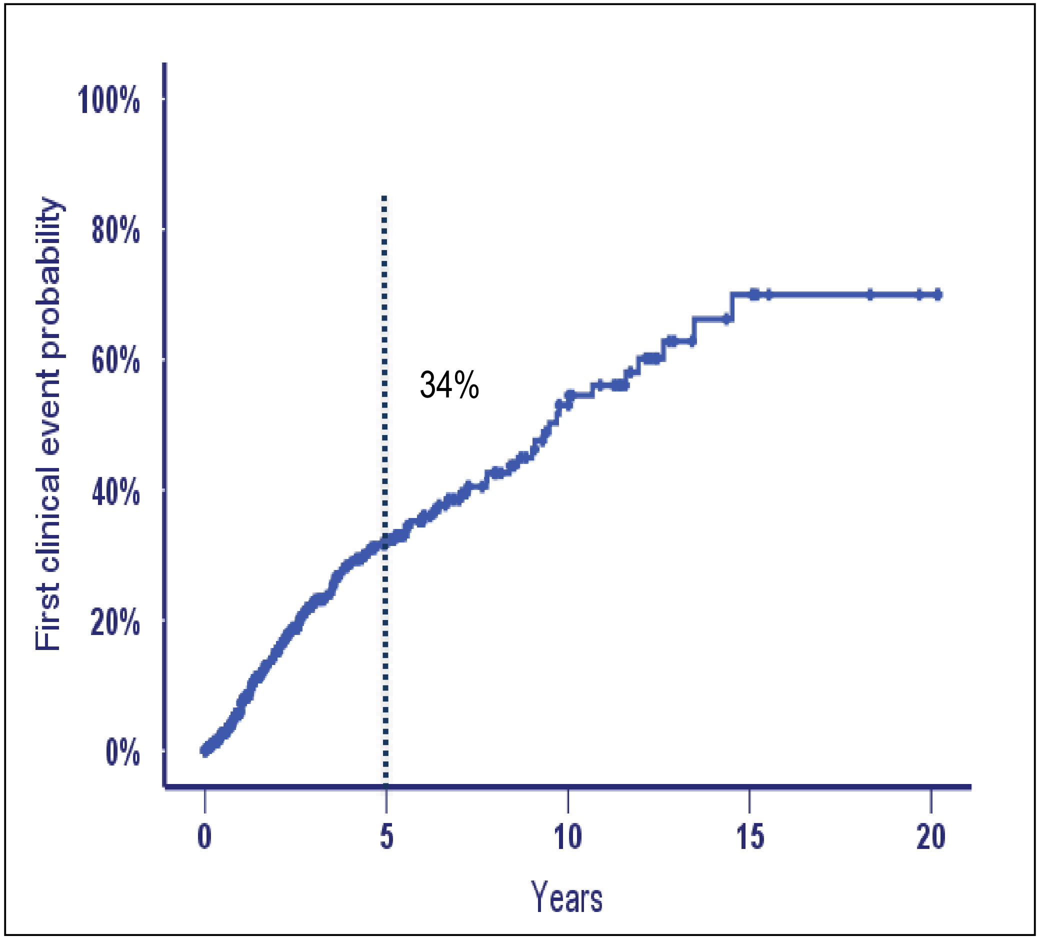
39y healthy control for a
Parkinson's disease study



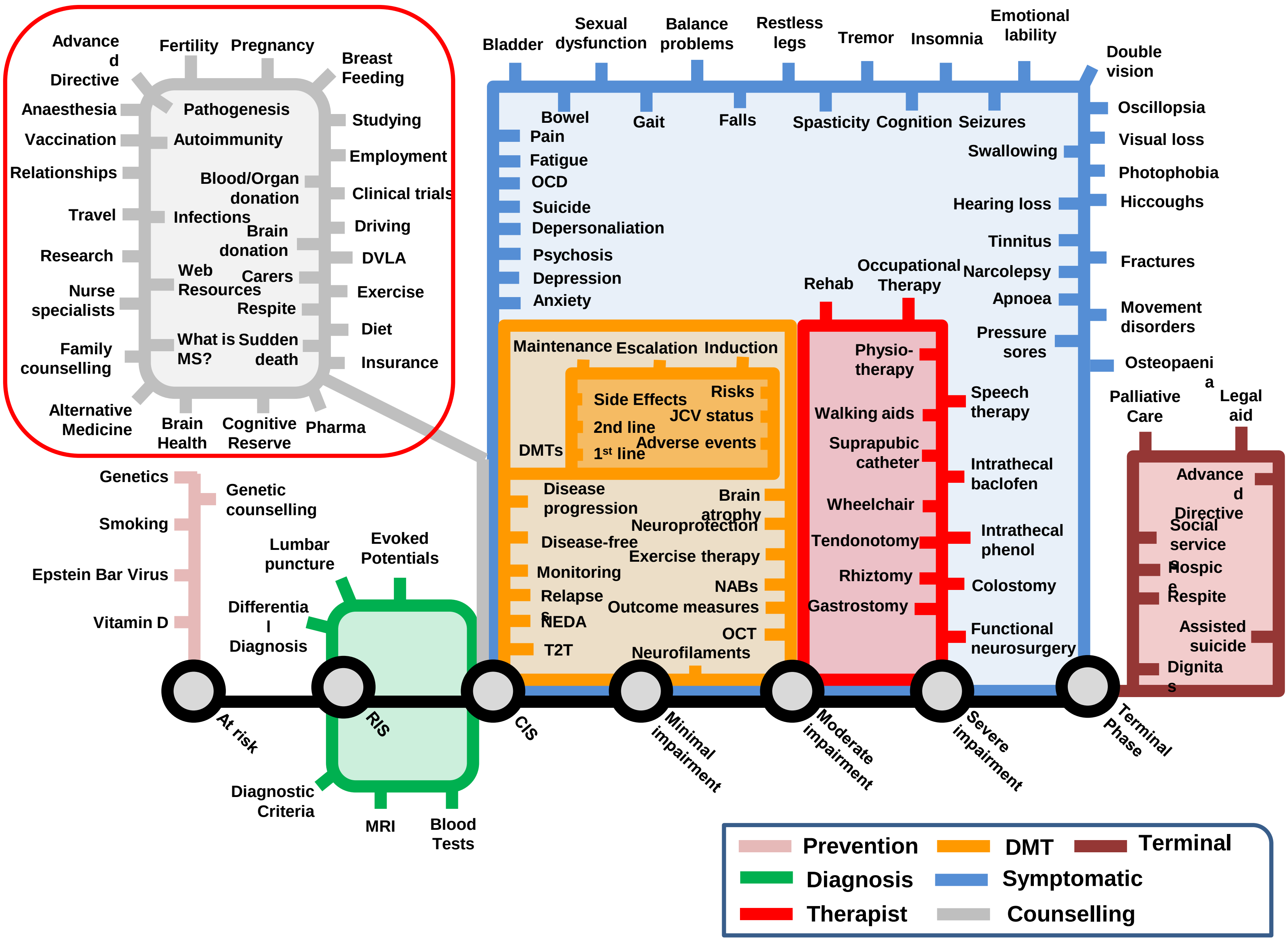



*McDonnell et al, Mult Scler 2003*

# Evolution to CDMS at 5-years



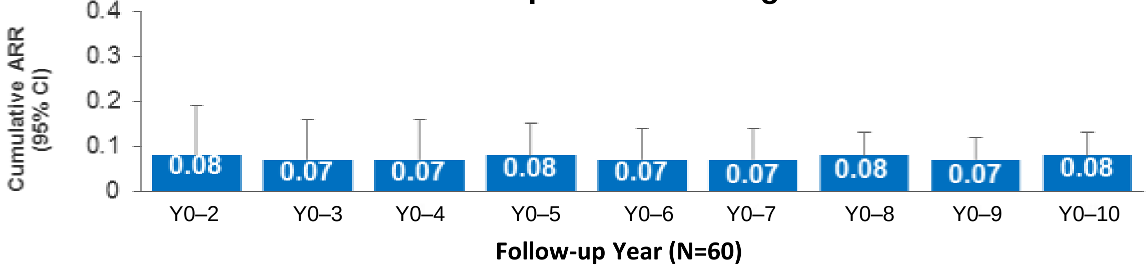

*Okuda et al. Neurology 2011;76:1–7*



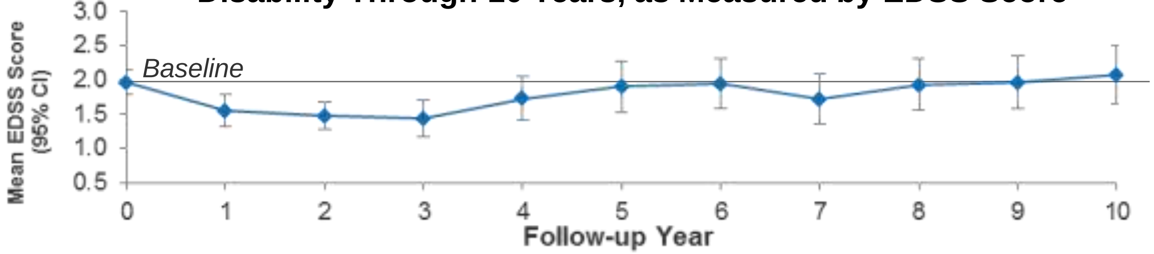

www.ms-res.org

# [P679] Efficacy and Safety of Alemtuzumab in Patients With RRMS is Durable Over 10 Years: Follow-up From the CAMMS223 Study (*Selected Data*)

**Annualized Relapse Rate Through 10 Years**




**Disability Through 10 Years, as Measured by EDSS Score**




| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **No. of Patients** | 60 | 60 | 59 | 59 | 45 | 48 | 48 | 54 | 57 | 56 | 54 |

- At CAMMS223 baseline, 100% of patients had Gd-enhancing lesions
  - After alemtuzumab treatment, the proportion of patients with Gd-enhancing lesions was low (CARE-MS extension baseline: 8.3%; CARE-MS extension Years 1, 2, and 3 were 5.6%, 13.0%, and 5.5%, respectively)

Selmaj K et al. ECTRIMS 2016, Poster P679.